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Yu-Hsi Chen

SUMMARY

- Senior CV Algorithm Engineer with 10 years of experience in spatio-temporal modeling and deep learning at Academia Sinica (born 1991).
- Expertise in Robust Perception: Specialized in object detection, tracking, and re-identification, with a focus on achieving high-accuracy recognition under adverse conditions.
- Generative AI & Synthetic Data: Developed LD-YOLOv7 by integrating Latent Diffusion models for weather-degradation simulation, reaching 95.05% accuracy in extreme rain and fog.
- Optimization & Data Association: Hands-on experience in implementing Graph-based association (e.g., Bipartite Matching via the Hungarian Method) for stable multi-object tracking.
- Foundation Models & VLM: Applied Vision-Language Models and LoRA fine-tuning to construct Universal Action Spaces; actively deploying SAM2 and Grounding DINO for unsupervised video segmentation.
- Efficient Implementation: Proficient in Python and PyTorch, with practical experience in Multi-GPU training and leveraging CUDA-accelerated toolkits to optimize large-scale video processing pipelines.

EDUCATION

Master of Science in Electrical Engineering

09 2013 – 08 2015

Lunghwa University of Science and Technology

Taipei, Taiwan

Graduated **1st in department** (Rank 1/19, GPA: 92.34/100).

Master's thesis: *Full-Frame Video Stabilization via SIFT Feature Matching*.

Received the Excellent Paper Award at IIHMSP 2014.

Bachelor of Science in Computer Information and Network Engineering

09 2009 – 07 2013

Lunghwa University of Science and Technology

Taipei, Taiwan

Focus areas: Verilog development for camera auto-exposure (AE) and auto-white balance (AWB); HDR imaging using MATLAB.

WORK EXPERIENCE

Computer Vision Engineer (Research Assistant)

07 2015 - Present

Academia Sinica (Taiwan), Institute of Information Science(I.I.S.)

Taipei

- Designed and implemented deep learning models for computer vision tasks using Python and PyTorch, including CNNs, C3D, Siamese Networks, Transformers, and custom extensions of YOLOv7 for adverse conditions.
- **NeighborTrack[1]**: Achieved top performance on LaSOT and held #1 on UAV123 until 2024.
- Focus areas: object detection/tracking, person re-identification, video stabilization, and recent advances in VLMs, generative vision models, and foundation model applications.

SELECTED PROJECTS

Deep Learning-based Animal Behavior Analysis [3]

08 2025

I.I.S. Research, Keywords: /S4/VST/V-JEPA/VideoCLIP/SAM2/GroundingDINO

- Developed a modular system for behavior analysis using vision-only, world, and vision-language models (VST, V-JEPA, VideoCLIP) with a Universal Action Space built from Kinetics-600.
- Applied structured state space models (S4/ViS4mer) for long-term temporal modeling, paired with a focal-loss-based key action selector to enhance focus on behavior-relevant segments.
- Investigated VLMs (SAM2, GroundingDINO) for unsupervised video object segmentation and foreground/background separation.
- Project page: <https://github.com/franktpmvu/Universal-Action-Space>

License Plate Recognition in Low Quality Images using Latent Diffusion [2]

04 2024

I.I.S. Research, Keywords: YOLOv7/Latent Diffusion/Synthetic Data/Adverse Weather Simulation

- Created a synthetic license plate generation pipeline with simulated weather degradation, specifically targeting adverse conditions such as rain and fog to improve recognition robustness without relying on real-world training data.
- Developed LD-YOLOv7, an enhanced detection framework that integrates latent diffusion to restore noisy intermediate features and boost robustness in extreme weather.
- Achieved 87.38% accuracy on the AOLP dataset (on par with real-data-trained models) and 35.19% on the proposed WLP benchmark, outperforming prior methods in synthetic-to-real generalization.
- Project page: <https://github.com/franktpmvu/License-Plate-Recognition-LD-YOLOv7>

NeighborTrack [1] (Single Object Tracking)

06 2023

I.I.S. Research, keywords: Single Object Tracking/Context Modeling

- Graph-Theoretic Association: Formulated data association as a Weighted Bipartite Matching problem, utilizing the Hungarian Algorithm to achieve globally optimal trajectory association and mitigate identity switches.
- Spatial Context Modeling: Developed NeighborTrack, a lightweight module that incorporates spatiotemporal context from neighboring tracklets to stabilize tracking in dense or occluded environments.
- Benchmark Excellence: Achieved 72.2% AUC on the LaSOT benchmark, surpassing competitive baselines.
- Project page: <https://github.com/franktpmvu/NeighborTrack>

[Early Work] Video Stabilization (Master's Thesis)

08 2014

keywords: SIFT/Content-Preserving Warping

- Developed a full-frame video stabilization system using SIFT feature matching and homography estimation for global motion extraction.
- Applied Kalman filtering and Gaussian smoothing to correct jitter while preserving intentional camera motion.
- Awarded Excellent Paper at IIHMSP 2014.

RESEARCH PUBLICATIONS

- [1] Yu-Hsi Chen et al. "NeighborTrack: Single Object Tracking by Bipartite Matching With Neighbor Tracklets and Its Applications to Sports". In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*. June 2023, pp. 5138–5147.
- [2] Yu-Hsi Chen et al. "License plate recognition in low quality image by using Latent Diffusion YOLOv7". In: *2024 IEEE Conference on Artificial Intelligence (CAI)*. 2024, pp. 831–838. DOI: [10.1109/CAI59869.2024.00156](https://doi.org/10.1109/CAI59869.2024.00156).
- [3] Yu-Hsi Chen et al. *Deep Learning-based Animal Behavior Analysis: Insights from Mouse Chronic Pain Models*. 2025. arXiv: [2508.05138](https://arxiv.org/abs/2508.05138) [cs.CV]. URL: <https://arxiv.org/abs/2508.05138>.

SKILLS

Programming Python, MATLAB, L^AT_EX.

Development Tool Docker, Git.

Foundation Models Latent Diffusion, VideoCLIP, V-JEPA, SAM2, GroundingDINO.

Expertise Visual Modeling (object tracking, behavior recognition, detection), Temporal Modeling (S4, ViS4mer, Vision Transformers like VST), Video Understanding, Generative Vision (diffusion-based models). Long-term Temporal Modeling (S4, ViS4mer).

Communication Chinese (native), English (basic professional use: reading papers), Japanese (intermediate, JLPT N3 certified).